EV Technology Course — Module 16: New and Emerging EV Technologies

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EV technology is changing faster than any other part of the auto industry. This module surveys the new technologies you will hear about (megawatt charging, 1,000 V platforms, axial-flux and magnet-free motors, steer-by-wire, 48 V low-voltage systems, giga-casting, wireless charging, vehicle-to-grid, solid-state and sodium-ion batteries) and, most importantly, how mature each one is. Status is as of October 2026; check sources for updates.

1. Learning objectives

- Explain the working principle of each major emerging EV technology.
- Classify technologies by maturity: in production, early or limited production, pilot or research.
- Estimate the engineering benefit of each with simple calculations.
- Assess the cost and manufacturing implications for a supplier or OEM.
- Separate real progress from marketing claims.

2. Fundamentals

- **Technology maturity** is often measured with Technology Readiness Levels (TRL 1–9) and Manufacturing Readiness Levels (MRL). A lab demo is not a production part.
- Most new EV technologies aim at one of five goals: **faster charging, lower cost, lower mass, higher efficiency, or less dependence on scarce materials**.
- Adoption depends on cost per unit benefit, supply chain readiness and regulation.

3. Detailed technical explanation

Faster charging and higher voltage

- **800 V platforms** (Hyundai E-GMP, Porsche, Kia, many Chinese EVs) halve current for the same power, reducing cable mass and heat.
- **1,000 V and megawatt charging:** BYD's Super e-Platform (announced 17 March 2025) uses 1,000 V, up to 1,000 A, and up to 1 MW "flash charging", claiming about 400 km of range in 5 minutes; it pairs with a 30,000 rpm motor and SiC devices rated up to 1,500 V. It needs special chargers with buffer storage.

- **Silicon carbide (SiC)** is now mainstream in 800 V inverters; **gallium nitride (GaN)** is entering on-board chargers and DC-DC converters for smaller, more efficient units.

New motors

- **Axial-flux motors:** flux runs parallel to the shaft; the motor is a short, wide "pancake" with very high power density. Mercedes-owned YASA motors are in series production in the Mercedes-AMG GT 4-Door Coupe (three motors), built in Berlin; a 2025 prototype reached about 42 kW/kg.
- **Electrically excited synchronous motors (EESM):** no permanent magnets; the rotor is magnetised with a winding fed through brushes or wireless exciters. Used by Renault for years and by BMW's Gen6 drive in the Neue Klasse (iX3, 2025).
- **Rare-earth-free or reduced magnets:** ferrite-assisted reluctance motors and heavy-rare-earth-free NdFeB grades, driven by China's April 2025 export-licensing curbs on rare earths.
- **India:** a ₹7,280 crore scheme for sintered rare-earth permanent magnets (6,000 tonnes per year capacity) was approved by the Cabinet on 26 November 2025.

Chassis and electrical

- **Steer-by-wire:** no mechanical column; Tesla Cybertruck (2023) and Lexus RZ (selected markets) are in production; needs redundant power, sensors and controllers (ASIL D).
- **48 V low-voltage system:** Tesla Cybertruck uses a 48 V architecture instead of 12 V, cutting harness current by 4× and resistive losses by 16× for the same power.
- **Zonal E/E and central compute:** covered in Module 10; enables software-defined vehicles.
- **Giga-casting:** large single aluminium castings replace dozens of stamped parts (Tesla, Xiaomi, Volvo, others). Saves parts and welding but needs huge presses and raises repair cost.

Charging and grid

- **Wireless (inductive) charging:** a ground pad transfers power to a vehicle coil; standard SAE J2954 at 3.7–11 kW. Premium OEMs (e.g. Porsche for the Cayenne Electric) offer 11 kW systems; efficiency is about 90%+ versus ~95% for plugs.
- **Vehicle-to-load (V2L), vehicle-to-home (V2H), vehicle-to-grid (V2G):** bidirectional on-board chargers and ISO 15118-20 communication allow cars to power appliances or support the grid. V2L is common (Hyundai, Kia, MG, Tata); V2G is in pilots and limited markets.
- **Automatic charging robots** and **battery swapping** for cars (Nio) are niche.

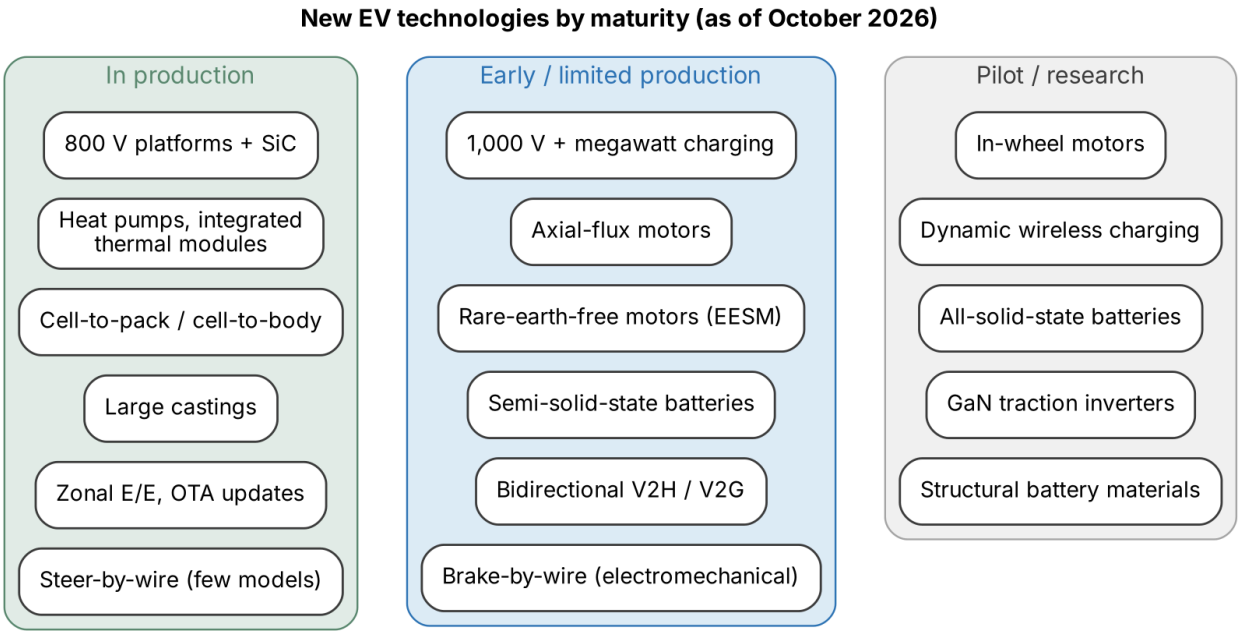
Batteries (linking to the battery course)

- **LMFP and high-manganese chemistries:** cheaper than NMC, more energy than LFP; in limited production.
- **Sodium-ion:** no lithium, good cold performance, lower energy density (~140–175 Wh/kg cell); CATL and BYD have production cells for small EVs and storage.
- **Semi-solid and all-solid-state:** semi-solid cells are in limited Chinese cars (e.g. MG, Nio); all-solid-state is in pilot lines, with 2027 expected as a small-batch vehicle-integration year and scale after 2030.
- **Cell-to-pack and cell-to-body:** fewer modules, better volumetric efficiency (BYD Blade, CATL Qilin, Tesla 4680 structural pack).

Maturity summary (October 2026)

Technology	Status
800 V, SiC inverters, cell-to-pack, V2L, giga-casting	In production (mainstream)
1,000 V / MW charging (BYD), axial-flux motors, EESM, steer-by-wire, 48 V LV, sodium-ion cells, semi-solid cells, wireless charging	Early or limited production
All-solid-state batteries, large-scale V2G, rare-earth-free motors at scale	Pilot or research

4. Diagram: new technologies by maturity



5. Step-by-step: evaluating a new technology as an engineer

1. Define the benefit: kg saved, Wh/km saved, minutes saved, ₹ saved, or risk removed.
2. Check maturity: TRL, MRL, who is in series production, volumes.
3. Estimate cost: material, process, capex, IP licensing, validation.
4. Check system impacts: thermal, safety, EMC, serviceability, repair.
5. Check supply chain: number of suppliers, localisation, raw material risk.
6. Run a pilot (prototype vehicle, limited series) with clear KPIs.
7. Decide adoption: now, next platform, or watch.

6. Engineering equations

Current at a given power and voltage:

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I = \frac{P}{V}
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Resistive loss:

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P_{loss} = I^2 R
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Average charging power to add range:

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P_{avg} = \frac{d \times c_{kWh/km}}{t}
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Power density:

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\rho_P = \frac{P_{peak}}{m_{motor}}
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7. Worked examples

Example 1: what 400 km in 5 minutes needs. 400 km at 0.15 kWh/km = 60 kWh. In 5 minutes (1/12 h): 60 × 12 = **720 kW average**, so peaks near 1 MW, as BYD claims.

Example 2: 400 V vs 800 V. At 350 kW charging, current is **875 A** at 400 V but **437.5 A** at 800 V. Cable loss (I^2R) falls to a quarter for the same cable.

Example 3: axial-flux power density. 550 kW from 13.1 kg = **42 kW/kg**, several times a typical radial-flux traction motor.

Example 4: wireless charging loss. Adding 50 kWh to the battery takes 52.6 kWh from the grid at 95% (plug) and 55.6 kWh at 90% (wireless): about **3 kWh more** per charge.

Example 5: solid-state mass saving. A 75 kWh pack needs 300 kg of cells at 250 Wh/kg but **187.5 kg** at 400 Wh/kg.

8. Real-world examples

- **BYD Han L / Tang L** launched with the Super e-Platform and a megawatt charging network rollout in China.
- **Mercedes-AMG GT 4-Door Coupe**: first series AMG with YASA axial-flux motors.
- **BMW iX3 (Neue Klasse)**: Gen6 EESM drive, 800 V, cylindrical cells.
- **Tesla Cybertruck**: steer-by-wire, 48 V LV architecture, rear-wheel steering.
- **India:** Tata, Mahindra and MG offer V2L; Indian firms are working on rare-earth magnet production and rare-earth-free motors; wireless charging and V2G remain at pilot stage.

9. Manufacturing applications

- **SiC:** wafer supply, die attach (sintering), module packaging; cost falling with 200 mm wafers.
- **Axial-flux:** precise rotor-stator air gap, soft magnetic composites or segmented laminations, new winding methods.
- **EESM:** rotor windings and brushes or contactless exciters; removes magnet supply risk.
- **Giga-casting:** 6,000–9,000 tonne presses, new alloys that do not need heat treatment, X-ray inspection.
- **Solid-state:** dry rooms, sulfide electrolytes sensitive to moisture, new stacking and pressing processes.
- **Costing:** each technology changes the cost model; e.g. EESM swaps magnet cost for copper, brushes and controller cost.

10. Common mistakes and misconceptions

- **"Solid-state EVs are already on sale everywhere."** Most "solid-state" cars today are semi-solid; all-solid-state is still in pilot stage.
- **"Megawatt charging works on any charger."** It needs special chargers, grid capacity and a battery designed for it.
- **"Axial flux is always better."** It is costlier and harder to manufacture; best for high-performance uses today.
- **"New means better value."** Mature technologies often give better cost per benefit.

11. Practical case study: magnet supply shock

Problem. In April 2025, China introduced export licences for several rare earths and magnets. Many Indian EV and two-wheeler makers faced magnet shortages within weeks.

Responses. Short term: inventory, alternate suppliers, redesigns with ferrite or heavy-rare-earth-free magnets. Medium term: EESM and reluctance motor development. Policy: India's ₹7,280 crore REPM scheme (November 2025).

Lesson. New technology choices are often driven by supply-chain risk, not only performance.

12. Revision notes

- 800 V and SiC are mainstream; 1,000 V / MW charging is early (BYD).
- Axial-flux (YASA/Mercedes) and EESM (BMW, Renault) are in series production.
- Steer-by-wire and 48 V LV are in limited production (Cybertruck, Lexus RZ).
- Semi-solid is in limited cars; all-solid-state is pilot, scale after 2030.
- Evaluate tech by benefit, maturity, cost, system impact, supply chain.

13. Glossary

Term	Meaning
TRL / MRL	Technology / manufacturing readiness level
SiC / GaN	Wide-bandgap semiconductor materials
Axial flux	Motor with flux parallel to the shaft
EESM	Electrically excited synchronous motor
V2L / V2H / V2G	Vehicle to load / home / grid
Giga-casting	Very large single-piece body casting
Semi-solid battery	Cell with reduced liquid electrolyte

14. Knowledge check

Conceptual

1. Why do higher voltages enable faster charging?
2. How does an EESM avoid rare-earth magnets?

3. What redundancy does steer-by-wire need?
4. Why is wireless charging less efficient?
5. What is the difference between semi-solid and all-solid-state batteries?

Numerical

1. Current at 1,000 kW and 1,000 V?
2. A 1.2 kW load at 12 V vs 48 V: currents and ratio of resistive loss?
3. Cell mass for 40 kWh using sodium-ion (160 Wh/kg) vs LFP (180 Wh/kg)?

Application

1. As a costing engineer, compare an EESM motor with a PMSM motor cost structure.
2. Recommend which three emerging technologies an Indian OEM should adopt first, and why.

15. Answer key

Conceptual

1. Same power with less current means smaller cables and lower heat, so higher power is practical.
2. The rotor magnetic field is created by a current-carrying winding instead of permanent magnets.
3. Redundant sensors, controllers, power supplies, communication and actuators, plus fail-operational design to ASIL D.
4. Magnetic coupling across an air gap and extra power conversion stages lose energy.
5. Semi-solid keeps some liquid or gel electrolyte; all-solid-state uses only solid electrolyte, which promises higher safety and energy density but is harder to make.

Numerical

1. 1,000,000 ÷ 1,000 = **1,000 A**.
2. 100 A at 12 V, 25 A at 48 V; loss ratio $(100/25)^2$ = **16× lower** at 48 V.
3. Sodium-ion **250 kg**; LFP **222 kg**.

Application

1. EESM removes magnet cost (and price/supply risk) but adds rotor copper, insulation, slip rings or wireless exciter, extra power electronics for field control and more complex rotor manufacturing; efficiency at light load can be higher, at low speed lower. Net cost depends on magnet prices.

2. Example: 800 V with SiC for faster charging (mature), reduced-rare-earth motors (supply security, REPM support), and V2L / connected features (low cost, customer value); watch solid-state and MW charging.

16. Further learning

- SMM solid-state battery reviews; BloombergNEF technology outlooks.
- IEEE Transactions on Transportation Electrification for motors and power electronics.
- SAE J2954 (wireless charging) and ISO 15118-20 (bidirectional charging) summaries.
- Munro teardowns of Cybertruck and new platforms.

Sources

- BYD unveils Super e-Platform with megawatt flash charging — BYD, 17 Mar 2025
- YASA industrialises axial-flux motor production under Mercedes-Benz — Automotive Manufacturing Solutions, 22 Jul 2026
- SMM Solid-State Battery H1 2026 Review — SMM, 2 Jul 2026
- Cabinet approves ₹7,280 crore rare-earth permanent magnets scheme — Tribune India, Nov 2025

Next: Module 17, capstone teardown and should-cost project.